

Tutorials (<https://www.sdss4.org/dr17/tutorials/>)

Download full files to my computer and write programs to use the data.	Science Archive Server (SAS)	The SAS houses parameter summary catalogs, spectra, and all intermediate data products in flat files within a directory structure. As described in Supplemental Analyses (/dr17/irspec/spectro_data_supplement/) there are multiple file trees in DR17 using different libraries. The default library is synspec with results in the synspec_rev1 (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/) filetree. (See Caveat (/dr17/irspec/caveats/#wronglsf) for details)	☆ allStar File (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allStar-dr17-synspec_rev1.fits) (4.0 Gb, datamodel (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStar.html)), but see also this caveat (/dr17/irspec/caveats#wronglsf)) ☆ allStarLite (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allStarLite-dr17-synspec_rev1.fits) (1.8 Gb, data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStarLite.html)) ☆ allVisit File (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allVisit-dr17-synspec_rev1.fits) (2.8 Gb, datamodel (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allVisit.html))
Use a Web-based GUI interface to select targets or view spectra.	Science Archive Webapp (SAW)	The SAW contains spectra and associated parameters in a searchable interactive interface. The SAW is loaded from the synspec_rev1 (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/) filetree.	☆ Infrared Spectrum Search (https://dr17.sdss.org/infrared/spectrum/search) ☆ Infrared Spectrum View Stars (https://dr17.sdss.org/infrared/spectrum/view/stars)
Query a database with SQL to define a sample.	Catalog Archive Server (CAS)	The CAS contains the information from the summary catalogs in a queryable database structure that can be accessed through the SkyServer or directly via SQL. The CAS is loaded from the synspec (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec/) filetree.	☆ IR Spec Query Form (http://skyserver.sdss.org/public/SearchTools/IRQS/) ☆ CasJobs SQL Query (http://skyserver.sdss.org/public/SearchTools/SQL/)

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Science Archive Server (SAS)

The entire spectroscopic dataset, at each stage of its data processing, is available through the SAS. For APOGEE, the entire set of visit and combined spectra can be found under the APOGEE_REDUX (<https://data.sdss.org/sas/dr17/apogee/spectro/redux/dr17/>) directory, as described by the datamodel (https://data.sdss.org/datamodel/files/APOGEE_REDUX/). The entire set of ASPCAP files can be found under the APOGEE_ASPCAP (<https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/>) directory (as described by the datamodel (https://data.sdss.org/datamodel/files/APOGEE_REDUX/)). It is possible to download files in bulk as described on the bulk downloads (/dr17/data_access/bulk) page.

DR17 does include several "supplemental analyses" for ASPCAP stellar parameters and abundances, for more information on these see the supplementary analysis page (/dr17/irspec/spectro_data_supplement).

In the subsections to follow, we describe some of the main data files and explain how to find these in the SAS.

Summary catalogs

The tables below give the summary catalog files in FITS format on the SAS. These are found in the following directory: </sas/dr17/apogee/spectro/aspcap/dr17/synspec> (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/) (but see also this caveat (</dr17/irspec/caveats#wronglsf>))

Summary Catalogs

Contents	File Name & Direct Link	Data Model
summative catalog containing all quantities for each combined spectrum	allStar-dr17-synspec_rev1.fits (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allStar-dr17-synspec_rev1.fits) (4.0 Gb) See also Caveat. (/dr17/irspec/caveats#wronglsf)	Data Model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VE_RS/ASPCAP_VERS/allStar.html)
"lite" summative catalog containing a reduced set of quantities for each combined spectrum	allStarLite-dr17-synspec_rev1.fits (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allStarLite-dr17-synspec_rev1.fits) (1.8 Gb)	Data Model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VE_RS/ASPCAP_VERS/allStarLite.html)

summative catalog containing quantities derived from each individual visit spectrum	allVisit-dr17-synspec_rev1.fits (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/allVisit-dr17-synspec_rev1.fits) (2.8 Gb)	Data Model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allVisit.html).
catalog of Cannon stellar parameters and abundances	Not produced in DR17; Please see DR15 Data Access Documentation (/dr15/irspec/spectro_data/) for details on how to retrieve these files.	DR15 Cannon Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/APSTAR_VERS-DR14/ASPCAP_VERS/RESULTS_VERS/CANNON_VERS/).

Spectral Data

Combined ASPCAP Spectrum:: aspcapStar

The pseudo-continuum normalized, combined spectrum, with best matching synthetic ASPCAP spectrum is recorded as one file per object in the aspcapStar file (data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/TELESCOPE/FIELD/aspcapStar.html)). These are sorted by telescope and field, and can be found in the SAS using a path constructed as follows:

```
/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/TELESCOPE/FIELD/
```

The file name is constructed as:

```
aspcapStar-dr17-APOGEE_ID.fits
```

Where TELESCOPE , FIELD and APOGEE_ID are tags for each object that can be found in the allStar (or equivalent) summary file. Thus, to acquire this file, the user can build the URL from information in the allStar file.

Combined Spectrum:: apStar or asStar

The combined, RV-corrected spectra are recorded as one file per object in the apStar or asStar files, where apStar files are created for objects observed by the northern spectrograph and asStar files are created for objects observed by the southern spectrograph. These are sorted by telescope and field, and can be found in the SAS using a path constructed as follows:

/sas/dr17/apogee/spectro/redux/dr17/stars/TELESCOPE/FIELD

The filename is constructed as:

apStar-dr17-APOGEE_ID.fits, asStar-dr17-APOGEE_ID.fits

In the allStar (or equivalent) summary file, the filename for the combined spectrum is contained in the `FILE` tag. Thus, the user can build the URL from information contained in the allStar file.

Visit Spectrum:: apVisit or asVisit

The individual visit spectra are recorded as one file per visit in the apVisit or asVisit files. These are sorted by telescope, plate, MJD , and FIBERID . These files can be found in the SAS using a path constructed as follows:

/sas/dr17/apogee/spectro/redux/dr17/visit/TELESCOPE/FIELD/PLATE/MJD

The filename is constructed as:

apVisit-dr17-PLATE-MJD-FIBERID.fits or asVisit-dr17-PLATE-MJD-FIBERID.fits

This information can be acquired from the allVisit summary file, or in the VISITS tag of the allStar file. The `VISITS` tag contains a comma-separated list of the visit file names (without `.fits` appended).

Summary Table for Spectral Data

APOGEE File Name	Contents	SAS Path	Data Model
aspcapStar	Pseudo-continuum normalized, combined spectrum, with best matching synthetic ASPCAP spectrum, one file per object	/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/TELESCOPE/FIELD/aspcapStar-dr17-APOGEE_ID.fits	Data Model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/TELESCOPE/FIELD/aspcapStar.html)
apStar/asStar	Combined spectrum, one file per object	/sas/dr17/apogee/spectro/redux/dr17/stars/TELESCOPE/FIELD/FILE	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/stars/TELESCOPE/FIELD/apStar.html)
apVisit/asVisit	Individual visit spectrum, one file per visit	/sas/dr17/apogee/spectro/redux/dr17/visit/TELESCOPE/FIELD/PLATE/MJD/filename	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/visit/TELESCOPE/FIELD/PLATE_ID/MJD5/apVisit.html)

Intermediate Data Products

In addition to the summary catalogs and spectral data, all intermediate data products are also provided in the data release. Here the process is presented in the order of operations from targeting to final ASPCAP parameters and abundances. The pipeline code is available on github: sdss/apogee (<https://github.com/sdss/apogee>).

Targeting Field Summary Files

DR17 includes four types of files containing useful information on the stars (both targeted and non-targeted) in APOGEE fields and on the fields themselves, the designs, and the plates in which the targets are organized (see further details on this organizational structure in Zasowski et al. 2013 (<http://adsabs.harvard.edu/abs/2013AJ....146...81Z>) and Zasowski et al. 2017 (<http://adsabs.harvard.edu/abs/2017AJ....154..198Z>)).

All summary files can be found in this directory: `/sas/dr17/apogee/target/` (<https://data.sdss.org/sas/dr17/apogee/target/>)

Targeting Field Summary Files

Summary File	Description	Data Model
apogee2Field/apogeeField	Summary file that contains information for all fields, such as central coordinates	Data Model (https://data.sdss.org/datamodel/files/APOGEE_TARGET/apogee2Field.html)
apogee2Design/apogeeDesign	contains information on the individual designs, or groups of stars, like color limits and cohort fiber allocations	Data Model (https://data.sdss.org/datamodel/files/APOGEE_TARGET/apogee2Design.html)
apogee2Plate/apogeePlate	contains information on the physical plates, such as drill angle	Data Model (https://data.sdss.org/datamodel/files/APOGEE_TARGET/apogee2Plate.html)

In addition to these summary files, we also have the targeting input files that are used to design each plate. These are organized by survey and by field. They are contained in folders within `/sas/dr17/apogee/target/` (<https://data.sdss.org/sas/dr17/apogee/target/>) and the links provided below go to the proper sub-directory.

Targeting Input Files

Field File	Description	Data Model
apogee2Object	contains photometric and proper motion information on the candidate targets in the APOGEE-2 fields; one file for each field in the survey.	Data Model (http://data.sdss.org/datamodel/files/APOGEE_TARGET/apogee2Object/apogee2Object.html)
apogeeObject	contains photometric and proper motion information on the candidate targets in the APOGEE-1 fields; one file for each field in the survey.	Data Model (http://data.sdss.org/datamodel/files/APOGEE_TARGET/apogeeObject/apogeeObject.html)
apogee1mObject	contains photometric and proper motion information on the candidate targets in the APOGEE 1m fields; one file for each field in the survey.	Similar to apogeeObject (http://data.sdss.org/datamodel/files/APOGEE_TARGET/apogeeObject/apogeeObject.html).

Observation Files

APOGEE raw image data are also available on the SAS. These are organized by telescope and MJD . To find these files, the user needs to have acquired the MJD for each visit in the allVisit file.

Files Generated during Data Collection

TELESCOPE	Telescope	Location of Files
apo25m	Apache Point Observatory 2.5 meter	apogee/spectro/data/MJD (https://data.sdss.org/sas/dr17/apogee/spectro/data/).
lco25m	Las Campanas Observatory 2.5 meter	apogee/spectro/data2s/MJD (https://data.sdss.org/sas/dr17/apogee/spectro/data2s/).
apo1m	Apache Point Observatory 1.0 meter	apogee/spectro/data1m/MJD (https://data.sdss.org/sas/dr17/apogee/spectro/data1m/).

The raw data files are named apR- CHIP - EXPOSURE .apz for data taken by the northern spectrograph and asR- CHIP - EXPOSURE .apz for data taken by the southern spectrograph. The CHIP can be “a”, “b”, or “c”, where these letters correspond to the “red”, “green”, and “blue” APOGEE detectors, respectively.

The intermediate-level ap2D and ap1D files are available, but users should consult the data models (ap1D (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/exposures/INSTRUMENT/MJD5/ap1D.html) and ap2D (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/exposures/INSTRUMENT/MJD5/ap2D.html)), as well as the APOGEE reduction paper (Nidever et al. 2015 (<https://ui.adsabs.harvard.edu/abs/2015AJ....150..173N/abstract>))) before using these files.

Visit Level Processing

The visit-level processing occurs in a code called apred and is described on Visit Reduction. These are organized by telescope, FIELD, PLATE, and MJD. The file path is constructed as follows:

```
/sas/dr17/apogee/spectro/redux/dr17/visit/TELESCOPE/FIELD/PLATE/MJD/
```

Files used or created by apred for Visit Processing

File	input or output	File Description	Data Model
apPlan/asPlan	input	the apred processing plan	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/visit/TELESCOPE/FIELD/PLATE_ID/MJD5/apPlan.html).
apCFrame/asCFrame	output	calibrated spectra file for each chip of each exposure	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/visit/TELESCOPE/FIELD/PLATE_ID/MJD5/apCframe.html).

apPlate/asPlate	output	Summary file for all exposures of each chip containing the 300 combined flux- and wavelength-calibrated spectra	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/visit/TELESCOPE/FIELD/PLATE_ID/MJD5/apPlate.html)
apPlateSum/asPlateSum	output	Summary file with metadata for the objects with apPlate/asPlate files	
apVisit/asVisit	output	for a given object on a given plate/MJD the combined flux- and wavelength-calibrated spectra for all exposures	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/visit/TELESCOPE/FIELD/PLATE_ID/MJD5/apVisit.html)
apVisitSum	output	Summary file with the derived parameters for objects with apVisit/asVisit files (no spectra)	

Combined Spectra

The visit spectra for each target are combined as described on the visit combination page. This occurs in the following directory:

```
/sas/dr17/apogee/spectro/redux/dr17/stars/TELESCOPE/FIELD
```

Files Created for Visit Combination

File	input or output	File Description	Data Model
apStar/asStar	output	Combined spectrum for an individual object.	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/stars/TELESCOPE/FIELD/apStar.html)
apStarLSF	output	one for each object	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/stars/TELESCOPE/FIELD/apStarLSF.html)
apField/asField	output	Radial velocities and target information for all stars in a given field	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/stars/TELESCOPE/FIELD/apField.html)
apFieldVisits	output	Table of individual visit RVs for a given field	Data Model (https://data.sdss.org/datamodel/files/APOGEE_REDUX/APRED_VERS/stars/TELESCOPE/FIELD/apFieldVisits.html)

APOGEE Stellar Parameters and Chemical Abundances Pipeline (ASPCAP)

The ASPCAP stellar parameters pipeline produces one intermediate file for each field that is then merged to form the allStar summary file. These “aspcapField” files are found in the aspcap directories sorted by telescope and field.

```
/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/TELESCOPE/FIELD/aspcapField-FIELD.fits
```

Files created by ASPCAP

File	input or output	File Description	Data Model
aspcapField	output	ASPCAP stellar parameters (logg, Teff, [M/H], etc.), normalized spectrum, and best fit spectrum for each stars in a given field.	Data Model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/TELESCOPE/FIELD/aspcapField.html).

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DR17 Value Added Catalogs

Note: All VACs were produced with the results synspec

(<https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec/>) filetree, whereas we recommend use of the synspec_rev1 (https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/) filetree. (See this Caveat ([/dr17/irspec/caveats/#wronglsf](#))).

The following Value Added Catalogs (VACs) are available for the DR17 datasets:

- ☆ Open Clusters (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=open-cluster-chemical-abundances-and-mapping-catalog) -- Provides membership analysis for stars in the vicinity of known open clusters
- ☆ The Joker (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=orbital-parameter-samplings-of-apogee-2-stars-from-the-joker) -- Provides stellar orbital parameters from modeling multi-epoch RV measurements using The Joker
- ☆ SB2 (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=double-lined-spectroscopic-binaries-in-apogee) -- Provides spectroscopic binary identifications
- ☆ APOGEE NET (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=apogee-net:-a-unified-spectral-model) -- Provides an independent analysis of APOGEE spectra using neural nets
- ☆ Red Clump ([https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=apogee-red-clump-\(rc\)-catalog](https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=apogee-red-clump-(rc)-catalog)) -- Provides a catalog of red clump stars using spectro-photometric selections.
- ☆ AstroNN (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=the-astronn-catalog-of-abundances,-distances,-and-ages-for-apogee-dr17-stars) -- Provides stellar parameters, elemental abundances, distances, and ages from a neural network-based code trained on the spectra; orbital parameters are provided using these distances and GalPy.
- ☆ StarHorse (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=apogee-dr17-starhorse-distances,-extinctions,-and-stellar-parameters) -- Provides distances and extinctions using spectro-photometry for all stars in DR17
- ☆ APOGEE-Centric Fire Simulations (https://www.sdss4.org/dr17/data_access/value-added-catalogs/?vac_id=apogee-on-fire-simulation-mocks) -- Provides mock catalogs with APOGEE-error model using the Ananke framework to sample the Latte Simulations

Additional Value Added Catalogs will be made public in 2022.

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Catalog Archive Server (CAS)

Note: The CAS is loaded with the results in the synspec

(<https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec/>) filetree. (See this Caveat (</dr17/irspec/caveats/#wronglsf>)).

APOGEE target information and derived radial velocities, stellar parameters, and elemental abundances are loaded into the Catalog Archive Server (CAS) database. CAS contains a number of ways of interfacing with the database, including SQL-based access through the CasJobs (<http://skyserver.sdss.org/Casjobs/>) portal that allows you to save and analyze all your search results.

The SkyServer Explore tool (<https://skyserver.sdss.org/public/VisualTools/explore/summary>) provides a quick way to find APOGEE spectra for the desired object. The QuickLook pane shows an image and spectrum for each SDSS object. For objects with APOGEE spectra, the tool includes an APOGEE section. The APOGEE section of the Explore tool shows a composite spectrum, radial velocities, and derived stellar atmospheric parameters, while also providing links to additional data, including the spectrum as a FITS file.

All APOGEE catalog data are available through the search tools of SkyServer. The Infrared Spectroscopy Query Form (<http://skyserver.sdss.org/public/SearchTools/IRQS/>) lets you search for APOGEE catalog objects by position, spectral classification, velocity, and other constraints from spectroscopic and/or photometric parameters. SQL Search (<http://skyserver.sdss.org/public/SearchTools/SQL/>) lets you create your own search using the SQL database programming language; see the SkyServer SQL Tutorial (<https://skyserver.sdss.org/dr16/en/help/howto/search/searchhowtohome.aspx>) to learn how to write SQL queries.

We provide some use-cases for the CAS tools on Examples (</dr17/irspec/catalogs/#CAS>).

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Science Archive Webapp (SAW)

Note: The SAW is loaded results from the synspec_rev1

(https://data.sdss.org/sas/dr17/apogee/spectro/aspcap/dr17/synspec_rev1/) filetree. (See this Caveat (</dr17/irspec/caveats/#wronglsf>)).

The Science Archive Webapp (SAW) provides a searchable interface for infrared spectra and includes both an interactive view of the spectra and a data download facility (supports both rsync and wget):

- ☆ Infrared Plate Search (<https://dr17.sdss.org/infrared/plate/search>)
- ☆ Infrared Spectrum Search (<https://dr17.sdss.org/infrared/spectrum/search>)
- ☆ Infrared Spectrum View Visits (<https://dr17.sdss.org/infrared/spectrum/view/visits>)
- ☆ Infrared Spectrum View Stars (<https://dr17.sdss.org/infrared/spectrum/view/stars>)

The SAW permits some simple object searches as well as interactively plots their associated spectra. It can also be used to construct lists of spectra to be downloaded in bulk; see further instructions in Bulk Data Downloads (/dr17/data_access/bulk/).

We provide some use-cases for the SAW tools on Examples (</dr17/irspec/catalogs/>).

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Backward Compatibility

The DR16 release involved a major reorganization of the SAS file structure for APOGEE data that is maintained in DR17. For consistency with previous data releases, we provide the equivalent pipeline versions here. Users wishing to access data from data releases are advised to refer to those pages, e.g., DR15 Data Access (/dr15/irspec/spectro_data/).

DR17 Pipeline Versions

APOGEE tags each stage of its pipeline with specific version numbers that are then included in the file names and as tags in the allStar (or equivalent) summary file. DR17 uses the following reduction/configuration versions:

- ☆ **APRED_VERSION::** dr17
- ☆ **APSTAR_VERSION::** stars
- ☆ **ASPCAP_VERSION::** synspec
- ☆ **RESULTS_VERSION::** OBSOLETE

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APOGEE Overview (<https://www.sdss4.org/dr17/irspec/>)

APOGEE DR17 Synopsis (https://www.sdss4.org/dr17/irspec/dr_synopsis/)

APOGEE DR17 Data Access (https://www.sdss4.org/dr17/irspec/spectro_data/)

Examples (<https://www.sdss4.org/dr17/irspec/catalogs/>)

Tutorials (<https://www.sdss4.org/dr17/irspec/apogee-tutorials/>)

Using APOGEE Data

Radial Velocities (<https://www.sdss4.org/dr17/irspec/use-radial-velocities/>)

Stellar Parameters (<https://www.sdss4.org/dr17/irspec/parameters/>)

Stellar Abundances (<https://www.sdss4.org/dr17/irspec/abundances/>)

Spectra (<https://www.sdss4.org/dr17/irspec/spectra/>)

Targets/Samples (<https://www.sdss4.org/dr17/irspec/targettingbits/>)

Caveats (<https://www.sdss4.org/dr17/irspec/caveats/>)

Technical Papers (<https://www.sdss4.org/dr17/irspec/technical-papers-and-additional-references/>)

Targeting Information (<https://www.sdss4.org/dr17/irspec/targets/>)

Selection Biases (<https://www.sdss4.org/dr17/irspec/targets/selection-biases/>)

Special Programs ▾ (<https://www.sdss4.org/dr17/irspec/targets/special-programs/>)

Observations & Reductions:

Observations (<https://www.sdss4.org/dr17/irspec/apogee-observing/>)

Visit Reduction (<https://www.sdss4.org/dr17/irspec/apred/>)

Visit Combination (https://www.sdss4.org/dr17/irspec/spectral_combination/)

Radial Velocities (<https://www.sdss4.org/dr17/irspec/radialvelocities/>)

Parameter & Abundance Determinations:

ASPCAP (<https://www.sdss4.org/dr17/irspec/aspcap/>)

Spectral Libraries (<https://www.sdss4.org/dr17/irspec/apogee-libraries/>)

APOGEE Bitmasks (<https://www.sdss4.org/dr17/irspec/apogee-bitmasks/>)

Explore

[Data Release 17 \(/dr17/\)](/dr17/)

[DR17 Data Access \(/dr17/data_access/\)](/dr17/data_access/)

[DR17 Scope \(/dr17/scope/\)](/dr17/scope/)

[DR17 Tutorials \(/dr17/tutorials/\)](/dr17/tutorials/)

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About

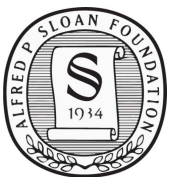
[How to Cite SDSS \(/collaboration/citing-sdss/\)](/collaboration/citing-sdss/)

[Image Use Policy \(/collaboration/#image-use\)](/collaboration/#image-use)

[Publication Policy \(/collaboration/#publication-policy\)](/collaboration/#publication-policy)

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SDSS is supported by



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Funding for SDSS is provided by the Alfred P. Sloan Foundation, the Participating Institutions, and the U.S. Department of Energy Office of Science.